

Survey Research and Design in Psychology

Tutorial 4 - Multiple linear regression

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- ▶ Talk about the general steps for multiple linear regression analyses
- ▶ Conduct multiple linear regression analyses
- ▶ Sample write ups

Steps

- 1) Draw your model - identify your independent and dependent variables
- 2) Check the assumptions
- 3) Choose the type of multiple linear regression and run it
- 4) Interpret the output
- 5) If meaningful, write the regression equation

Drawing a model

- ▶ Use a path diagram
- ▶ Create a venn diagram (note - this is harder with more predictors!)

Assumptions of multiple linear regression

- ▶ Level of measurement
- ▶ Sample size
- ▶ Normality
- ▶ Linearity
- ▶ Homoscedasticity
- ▶ Multicollinearity
- ▶ Residuals are normally distributed

Types of multiple linear regression

- ▶ Standard
- ▶ Hierarchical
- ▶ Stepwise/Forward/Backward

Example 1 - Standard multiple linear regression

- ▶ Open **Tutorial 4 - dataset 1**
- ▶ Deterrence theory = whether or not one obeys the law depends on how likely they think punishment for disobeying is
- ▶ Do certainty and severity predict compliance?
 - ▶ Compliance with the law $\sim$ certainty of getting caught + severity of consequences

Remember - Sample size

- ▶ Enough data is needed to provide reliable estimates of the correlations between variables
- ▶ $N > 50$ cases + $N > 10$ to 20 cases x number of independent variables, otherwise the estimates of the regression line are probably unstable and are unlikely to replicate if the study is repeated
- ▶ Green (1991) and Tabachnick & Fidell (2013) suggest:
 - ▶ $50 + 8(k)$ for testing an overall regression model and
 - ▶ $104 + k$ when testing individual predictors (where K is the number of independent variables)
 - ▶ Based on detecting a medium effect size ($\beta \geq .20$), with a critical $\alpha \leq .05$, with power of 80%
- ▶ Overall - have 10 to 20 times as many observations as independent variables

Check descriptive statistics

- ▶ Analyses > Exploration > Descriptives
- ▶ Move all variables across and ask for the defaults, as well as:
 - ▶ Statistics:
 - ▶ Skewness
 - ▶ Kurtosis
 - ▶ Plots:
 - ▶ Histogram

- ▶ Request a correlation matrix with compliance, certainty and severity
- ▶ Analyses > Regression > Correlation matrix > move variables of interest across > Plot - correlation matrix
 - ▶ Are our variables related?
 - ▶ Are there linear relations between independent variables and dependent variables?

Regression

- ▶ Do certainty and severity predict compliance?
- ▶ Analyses > Regression > Linear regression
 - ▶ Dependent variable: compliance
 - ▶ Independent variable: certainty and severity

Assumption Checks

Assumption Checks

☐ Autocorrelation test
☒ Collinearity statistics
☐ Normality test
☒ Q-Q plot of residuals
☒ Residual plots

Data Summary

☒ Cook's distance

Model Fit

Fit Measures

☒ R
☒ R²
☒ Adjusted R²
☐ AIC
☐ BIC
☐ RMSE

Overall Model Test

☒ F test

Model Coefficients

Omnibus Test

☐ ANOVA test

Standardized Estimate

☒ Standardized estimate
☒ Confidence interval
Interval 95 %

Estimate

☐ Confidence interval
Interval 95 %

- ▶ Interpret model fit measures
- ▶ Regression coefficients
 - ▶ If certainty and severity are zero, what's the rate of compliance?
 - ▶ From this output, what values go into a regression equation?
 - ▶ Which is the stronger predictor?
- ▶ Cook's distance
- ▶ Q-Q plot and fitted vs residuals

What did we do and find?

- ▶ What type of regression did we do?
- ▶ Assumptions - were we okay on sample size, linearity, multicollinearity, distribution of residuals?
- ▶ Were there any multivariate outliers?
- ▶ Findings - how well did our independent variables predict or explain our dependent variable

Write up should include

- ▶ Amount of variance explained
 - ▶ R , R^2 and adjusted R^2
 - ▶ R^2 change at each step if a hierarchical multiple linear regression is being conducted
 - ▶ Inferential tests ($F(df)$, p)
- ▶ Significance, size, direction and relative contribution of each independent variable
- ▶ A table that includes:
 - ▶ Correlations
 - ▶ Beta for intercept and independent variables
 - ▶ Beta (β)
 - ▶ Statistical significance (t , p -value)
 - ▶ Explain the direction and size of the results
- ▶ Consider the percentage of explained variance

We were interested in the extent to which severity and certainty of punishment predicts increased compliance with the law. In the zero-order correlations presented in Table 1, the two predictors are moderately correlated with each other (without violating the assumption of multicollinearity), though not significantly related to Compliance. Compliance with the law was regressed onto severity and certainty using simultaneous multiple linear regression. This analysis was appropriate as we had sufficient sample size, variables were related in a linear fashion based on examination of scatterplots, Cook's Distance did not suggest the presence of any multivariate outliers, and the Q-Q plot indicates residuals are approximately normally distributed. The predictors explained 4% of the variance in the DV ($R = .20$, $R^2 = .04$, adjusted $R^2 = .007$; $F(2, 59) = 1.23$, $p = .300$), and the overall model was not significant. Neither certainty ($B = 0.15$, $\beta = .19$, 95% CI $[-.08, .46]$, $t(59) = 1.42$, $p = .160$) nor severity ($B = 0.02$, $\beta = .02$, 95%CI $[-.25, .29]$, $t(59) = .14$, $p = .887$) were significant predictors, indicating that any apparent relationship these have with compliance could be due to chance.

Example 2 - Hierarchical multiple linear regression

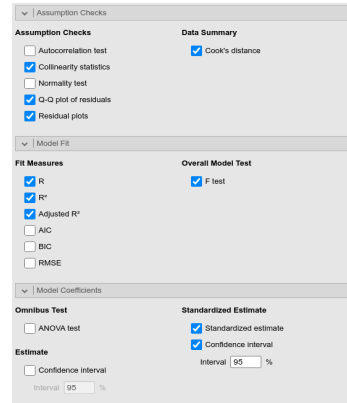
- ▶ We will now expand the deterrence theory model to include an additional predictor: perceived legitimacy of the authority
- ▶ Can adding legitimacy explain additional variance over and above certainty and severity?
 - ▶ Compliance with the law $\sim$ certainty of getting caught + severity of consequences
 - ▶ Compliance with the law $\sim$ certainty of getting caught + severity of consequences + legitimacy of authority

- ▶ Sample size = 62, predictors = 3. Is this enough?
 - ▶ Not really. The sample size could give us trustworthy estimates of the overall regression (i.e. R and R^2 values), but we might not get reliable estimates for individual predictors (B and β s)
 - ▶ Linearity, multicollinearity
 - ▶ Check the correlations and request a correlation matrix plot to examine the scatterplot

- ▶ Analyses > Regression > Linear regression > Model Builder
 - ▶ Dependent variable: compliance
 - ▶ Covariates: legitimacy, severity and certainty
 - ▶ Block 1: severity and certainty (Block 1)
 - ▶ Block 2: legitimacy (Block 2)

Assumptions and output

- ▶ Both remaining assumptions are tested while running the multiple linear regression
 - ▶ Multivariate outliers
 - ▶ Normality of residuals
- ▶ Note you can toggle between models



The screenshot displays a software interface for statistical analysis, organized into three main sections: Assumption Checks, Model Fit, and Model Coefficients.

Assumption Checks

Assumption Checks	Data Summary
☐ Autocorrelation test	☒ Cook's distance
☒ Collinearity statistics	
☐ Normality test	
☒ Q-Q plot of residuals	
☒ Residual plots	

Model Fit

Fit Measures	Overall Model Test
☒ R	☒ F test
☒ R ²	
☒ Adjusted R ²	
☐ AIC	
☐ BIC	
☐ RMSE	

Model Coefficients

Omnibus Test	Standardized Estimate
☐ ANOVA test	☒ Standardized estimate
	☒ Confidence interval
Estimate	Interval 95 %
☐ Confidence interval	
Interval 95 %	

What did we do and find?

- ▶ What type of regression did we do?
- ▶ Assumptions - were we okay on sample size, linearity, multicollinearity, distribution of residuals?
- ▶ Were there any multivariate outliers?
- ▶ Findings - how well did our independent variables predict or explain our dependent variable

Example write up

We tested the prediction that perceived legitimacy of the law would explain variance in compliance above and beyond certainty and severity of punishment using hierarchical multiple linear regression. Based on correlations between variables, legitimacy was strongly positively correlated with compliance ($r = .55$, $p < .001$), indicating that the more legitimate individuals perceive the law, the more they tend to comply. While legitimacy was also significantly correlated with certainty ($r = .39$, $p = .002$), this association was weak, and there was no significant relationship with severity ($r = .18$, $p = .160$), therefore we were not concerned about potential multicollinearity. Examinations of the scatterplots indicated that all relationships between variables were linear. Compliance with the law was regressed onto severity and certainty in Step 1 and Legitimacy in Step 2. Results of the MLR confirmed there were no issues with multivariate outliers (maximum CD = .17) and residuals approximated normal based on the Q-Q plot. . .

Example write up

The first step of the model did not fit the data well ($F(2, 59) = 1.23, p = .300$) and explained a nonsignificant, small portion of the variance in compliance (4.0%, $R = .20, R^2 = .04$, adjusted $R^2 = .01$). Adding legitimacy as a predictor in Step 2 explained an additional 26% of variance ($\Delta R^2 = .26, F(3, 58) = 21.20, p < .001$). Legitimacy was a significant positive predictor of compliance ($B = 0.55, \beta = .55, 95\% \text{ CI } [.31, .79], t(58) = 4.60, p < .001$). Based on non-overlapping confidence intervals, legitimacy was the significantly strongest predictor, and the association is interpreted to mean that greater perceived legitimacy predicts an increase in compliance. However, as the sample size for this analysis was low, the estimates for each predictor are potentially unreliable.

Example hierarchical regression table

	B	β	t	p
Step 1				
Intercept	2.97			
Certainty	0.15	.19 [-.08, .46]	1.42	.160
Severity	0.02	.02 [-.25, .29]	0.14	.887
Step 2				
Intercept	1.83			
Certainty	-0.01	-.01 [-.27, .22]	0.11	.909
Severity	-0.01	-.01 [-.25, .22]	0.10	.922
Legitimacy	0.55	.55 [.31, .79]	4.60	<.001

Example 3 - Standard multiple linear regression

- ▶ Open **Tutorial 4 - dataset 2**
- ▶ English achievement
 - ▶ Research question: to what extent can english achievement in year 8 be explained by parents socio-educational level, student's sex, amount of homework and attentiveness in class?
 - ▶ English achievement $\sim$ Sex + socio-education of parents + amount of homework + attentiveness in class

Questions to think about and do

- ▶ Think about level of measurement and get the descriptive statistics most appropriate for each variable
- ▶ Sample size
- ▶ Are our variables correlated. Are relationships linear?
- ▶ Run the regression
 - ▶ Choose the same options as before for assumption checks, model fit and standardised estimates
 - ▶ Evaluate the overall model
 - ▶ Which is the strongest predictor?
- ▶ Check for multivariate outliers
 - ▶ Cook's distance
- ▶ Normality of residuals
- ▶ Multicollinearity

What did we do and find?

- ▶ What type of regression did we do?
- ▶ Assumptions - were we okay on sample size, linearity, multicollinearity, distribution of residuals?
- ▶ Were there any multivariate outliers?
- ▶ Findings - how well did our independent variables predict or explain our dependent variable

Example write up

To assess the extent to which sex, parental socio-educational status, English homework and attentiveness in class predict English achievement, we first examined correlations between these variables (see Table 1). These suggested that students who pay more attention in class, do more homework, and were female achieved better outcomes in English, and confirmed that associations were linear (based on scatterplots), and predictors were not overly correlated with each other. Sample size was sufficient for simultaneous multiple linear regression, and our model indicated no issues with multivariate outliers (maximum CD = .05), with the residual plot suggesting a normal distribution. The overall model was significant and explained 21.9% of the variance in achievement ($R = .47$, $R^2 = .22$, adjusted $R^2 = .20$; $F(4, 144) = 10.10$, $p < .001$). Attentiveness in class was the only significant predictor, exhibiting a moderate positive relationship with English achievement such that those who were more attentive achieved better outcomes. The remaining predictor variables were not significant predictors and therefore any apparent relationship with the DV could have come about by chance (see Table 1 for regression coefficients).

Table 1

	1.	2.	3.	4.	B	β	t
Intercept					8.83		
1. Sex					0.62	.07 [-.10, .24]	0.80
2. Parental socio-education	-.02				0.12	.02 [-.12, .17]	0.31
3. English homework	.48***	-.03			0.70	.15 [-.02, .32]	1.77
4. Attentiveness	.20**	.06	.21**		1.53	.38*** [.22, .53]	4.84
5. English achievement	.22***	.05	.24***	.42***			

Note. *** $p < .001$, ** $p < .01$, * $p < .05$

Example 4 - Hierarchical multiple linear regression

- ▶ Can adding amount of homework and attentiveness in class explain additional variance over and above sex and socio-education of parents?
 - ▶ English achievement $\sim$ Sex + socio-education of parents
 - ▶ English achievement $\sim$ Sex + socio-education of parents + amount of homework + attentiveness in class

How would that write up change?

Details of the assumptions are identical, but the type of MLR is different (hierarchical), and there are two models to describe – predictors in the first step together explained 5.7% of the variance in English achievement ($R = .24$, $R^2 = .06$, adjusted $R^2 = .04$; $F(2, 146) = 4.43$, $p = .013$), with sex the only significant predictor, indicating achievement was higher among females. The second step explained an additional 16.2% of the variance ($\Delta R^2 = .16$; $F(4, 144) = 14.90$, $p < .001$) and revealed that with all predictors together, only attentiveness was significantly related to achievement, with more attentive students typically achieving better.

- ▶ What if assumptions are not met?

Sample size

- ▶ Enough data is needed to provide reliable estimates of the correlations
- ▶ If the sample size is small, we risk reporting findings that are unlikely to replicate if the study is repeated
- ▶ As a general rule, you need at least $50 + 8(k)$ for reliably testing an overall regression model and $104 + (k)$ for reliable estimates for individual predictors

- ▶ What if assumptions are not met?

Multivariate outliers

- ▶ Cook's distance (CD) analyses calculate a value for each participant (think of this as how likely they are to be a multivariate outlier)
- ▶ The output will give the maximum CD value
 - ▶ CD should not be greater than 1, so there may be outliers if the maximum CD is above 1
 - ▶ Some programs will show you the CD value for each participant, so you can then find all cases with a CD above 1 and investigate these, running the analyses with and without them to test whether they are influential outliers
 - ▶ For your lab report, comment on the likelihood of possible multivariate outliers based on the maximum CD value – it is either unlikely (max CD below 1) or possible that there is at least one multivariate outlier (max CD above 1)

- ▶ What if assumptions are not met?

Multicollinearity

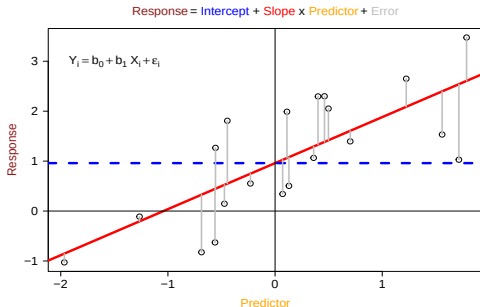
- ▶ Multicollinearity is potentially a problem if:
 - ▶ Correlation matrix shows large correlations ($> .7$) among any predictor variables
 - ▶ Tolerance values $< .3$
 - ▶ Variance Inflation Factor (VIF) > 3
 - ▶ Note: You only need to check one of these
- ▶ Reconsider the choice of predictor variables if multicollinearity is present

Troubleshooting

- ▶ What if assumptions are not met?

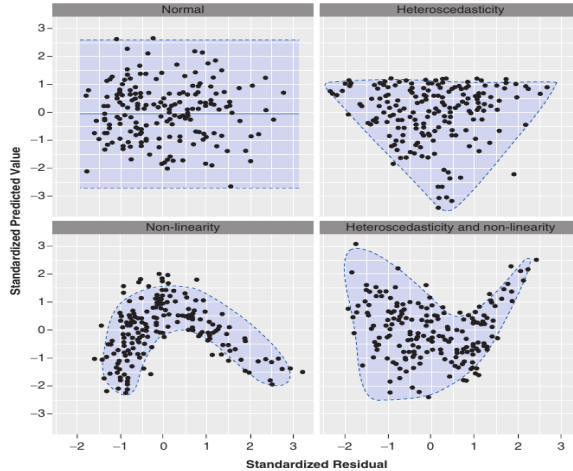
Normality of residuals

- ▶ Recall that residuals are the difference between predicted and observed scores, and these can be positive or negative.
- ▶ On average, the residuals = 0
- ▶ Error is random, and therefore residuals should be normally distributed around 0



- ▶ Remember that minor deviations are okay. It is unlikely that your Q-Q or Residual plot will look perfect, and we are only concerned if they show clear deviations from normality.
- ▶ What if the residuals are not normally distributed?
 - ▶ Try to reflect on why this might be the case
 - ▶ The most likely cause of non-normal distribution of residuals is non-normal variables
 - ▶ Include a statement indicating that the distribution of residuals is non-normal, and what this means – as violations of this assumption reduce the p-value, we might be more likely to make Type 1 errors.
 - ▶ Don't panic. You will not fail if the assumption is violated, but you may lose marks if you ignore this.
 - ▶ Reduce this risk of error by focusing on effect sizes and including confidence intervals (rather than focusing solely on p-values!)

Examples



- ▶ Analysis of residuals can pick up an outlier you missed in data screening, which is an easy fix. If this is the case, report the results after dealing with the out-of-range data.
- ▶ Analysis of residuals can highlight restriction of range on your variables. In this case, you should comment that the residuals plot shows some evidence of restriction of range, investigate which variable(s) in your regression might have restricted range. You should already know if this was a problem from looking at your univariate statistics. If not, check histograms of the constructs that went into your MLR. You can still report your MLR results (providing data are still normally distributed), but this is a limitation of your research.
- ▶ Analysis of residuals can pick up skewed data. Recall that an assumption of MLR is that continuous data is normally distributed, so severe skew or kurtosis violates this. If this is on one of your predictor variables and you still want to use it in the analysis, consider recoding it to a dichotomous variable.

Questions?

- ▶ You can ask questions any time on the discussion forums.
 - ▶ Jamovi
 - ▶ Tutorial